

Quantum Gravitation

Problem Set 12

Yuran Seo

Due 2026-11-22 23:59

Topic

Quantum States of Dense Media

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Derive the first displayed relation used in Quantum States of Dense Media. State all assumptions and conventions.

$$\omega_0^2(M_c) = 0, M_c = 1.9M_\odot$$

2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$\Psi = \langle a_{k \approx 0, s} \rangle / \sqrt{V}$$

3. Verify the third relation in two limiting regimes and explain the physical meaning of the result.

$$V_{eff} = a|\Psi|^2 + b|\Psi|^4 + \dots$$

Show enough intermediate work that another student can reproduce the calculation.